What are the cellular mechanisms through which hyaluronic acid-based treatments improve skin hydration and elasticity?

Research Paper By Parissa Luthra

Abstract

Hyaluronic acid (HA) is a natural substance already present in our skin. It has a special ability to hold a huge amount of water, which keeps the skin soft, smooth, and elastic. As we get older, the natural amount of HA in the skin decreases, leading to dryness and wrinkles. This is why many modern skin treatments use HA in different forms, such as creams, serums, skin boosters, and dermal fillers.

HA-based treatments work in several ways. First, HA fills the spaces around skin cells and acts like a sponge, holding water and reducing water loss. This keeps the skin barrier strong and makes the skin feel plump. Second, HA interacts with certain "sensor" proteins on skin cells. This wakes up fibroblasts—the cells that build collagen and elastin—which improves firmness and flexibility. Third, injectable HA not only adds volume but also supports these cells so they can produce new collagen over time. Finally, HA helps the skin use natural water channels (called aquaporins) that move water between cells, further improving hydration.

Together, these actions explain why HA treatments are effective: they trap water, activate skin-repairing cells, and strengthen the skin's barrier, resulting in healthier, more hydrated, and more elastic skin.

Keywords: hyaluronic acid; skin hydration; elasticity; collagen; fibroblasts; skin barrier; dermal fillers; water retention

Introduction

Hyaluronic acid (HA) is a natural sugar-like substance, also called a glycosaminoglycan, that is found in many parts of the human body, especially the skin. Its main job is to hold water in the spaces around skin cells, giving the skin a plump, soft, and springy feel. In fact, HA can hold up to

1,000 times its weight in water, which makes it one of the key molecules for keeping the skin well-hydrated and elastic. As people age, the natural levels of HA in the skin slowly decrease. This loss of moisture and structure is one of the reasons why older skin often looks drier, thinner, and less firm[1].

To understand how HA treatments work, it helps to look at the basic structure of the skin. The skin has two main layers: the epidermis (outer layer) and the dermis (deeper layer). The dermis contains collagen (which gives strength) and elastin (which gives stretch and flexibility). Both of these fibers sit in a gel-like material called the extracellular matrix (ECM). Hyaluronic acid is an important part of this matrix—it acts like a sponge that stores water, provides support, and helps skin cells communicate with each other. When HA levels fall, the skin's ability to stay hydrated and bounce back is reduced [2].

Because of its unique ability to hold water and support the skin's structure, hyaluronic acid (HA) is widely used in dermatology and cosmetic treatments. One common approach is the use of topical HA, which is found in creams and serums. These products help increase surface hydration, reduce dryness, and improve the overall feel of the skin. Clinical reviews show that HA-based skincare can noticeably improve skin moisture and reduce roughness[3]. Another major application is injectable HA, often referred to as dermal fillers. In this form, HA is chemically modified (cross-linked) to last longer once placed under the skin. Injectable HA is used to restore lost volume, smooth wrinkles, and improve skin firmness. Most side effects are mild and temporary, but complications may occur if the procedure is not performed correctly. In addition, researchers are developing advanced delivery systems such as nano-HA, liposomal HA, and microneedle patches. These technologies aim to help HA penetrate deeper skin layers and provide longer-lasting effects.

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[1] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923

[3]Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923)

Early clinical studies suggest promising results, but more evidence is needed before these methods become common practice (PMC).

HA treatments work in two main ways. On a physical level, HA binds water and reduces transepidermal water loss (TEWL), which strengthens the skin barrier and keeps it moist. On a biological level, HA can also act as a signaling molecule. It binds to cell surface proteins like CD44 and RHAMM, which then activate fibroblasts (the skin cells that make collagen and elastin). This helps repair the extracellular matrix and improves elasticity[4].

Another important pathway involves Aquaporin-3 (AQP3), a protein channel in the epidermis that moves water and glycerol between cells. Studies show that AQP3 supports skin hydration, barrier strength, and healthy cell turnover. By influencing AQP3 activity, HA treatments further improve moisture balance and smoothness. (Dicker, K. T., Gurski, L. A., Pradhan-Bhatt, S., Witt, R. L., Farach-Carson, M. C., & Jia, X. (2014). Hyaluronan: A simple polysaccharide with diverse biological functions. Acta Biomaterialia, 10(4), 1558–1570. https://doi.org/10.1016/j.actbio.2013.12.019)

The central research question of this paper is: ***What are the cellular mechanisms through which hyaluronic acid (HA)-based treatments improve skin hydration and elasticity?*** This study argues that HA improves skin health through a dual mechanism. On one hand, HA has a remarkable ability to hold water and strengthen the skin barrier, reducing dryness and maintaining moisture. On the other hand, it also acts as a signaling molecule that activates skin cells, stimulates fibroblasts to produce collagen and elastin, and regulates hydration channels such as aquaporin-3. Together, these effects explain why HA treatments have become a cornerstone of modern dermatology and anti-aging care(Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key

[4] (Misra, S., Hascall, V. C., Markwald, R. R., & Ghatak, S. (2015). Interactions between hyaluronan and its receptors (CD44, RHAMM) regulate the activities of inflammation and cancer. *Frontiers in Immunology*, *6*, 201. https://doi.org/10.3389/fimmu.2015.00201

molecule in skin aging. *Dermato-Endocrinology, 4*(3), 253–258. https://doi.org/10.4161/derm.21923))

I chose this topic because HA is one of the most widely used ingredients in skincare and cosmetic treatments today, yet many people are unaware of the science behind how it actually works. Understanding the cellular mechanisms gives a deeper appreciation of why HA is so effective and helps bridge the gap between cosmetic practice and scientific knowledge. It also highlights how biology and dermatology come together to address common concerns of aging, dryness, and skin health.

Skin Structure and Physiology Relevant to Hydration and Elasticity

To understand how hyaluronic acid (HA) treatments improve skin hydration and elasticity, it is first important to look at the basic structure of the skin and how it works. The skin is the largest organ of the body, and it is made of three main layers: the epidermis, the dermis, and the hypodermis. The epidermis is the outer layer that acts as a barrier, protecting us from water loss, bacteria, and environmental stress. The dermis lies just below and is much thicker; it contains blood vessels, nerves, and the connective tissue that gives skin its strength and flexibility. The hypodermis, the deepest layer, is made of fat and connective tissue that cushions the body and helps regulate temperature (MedlinePlus).

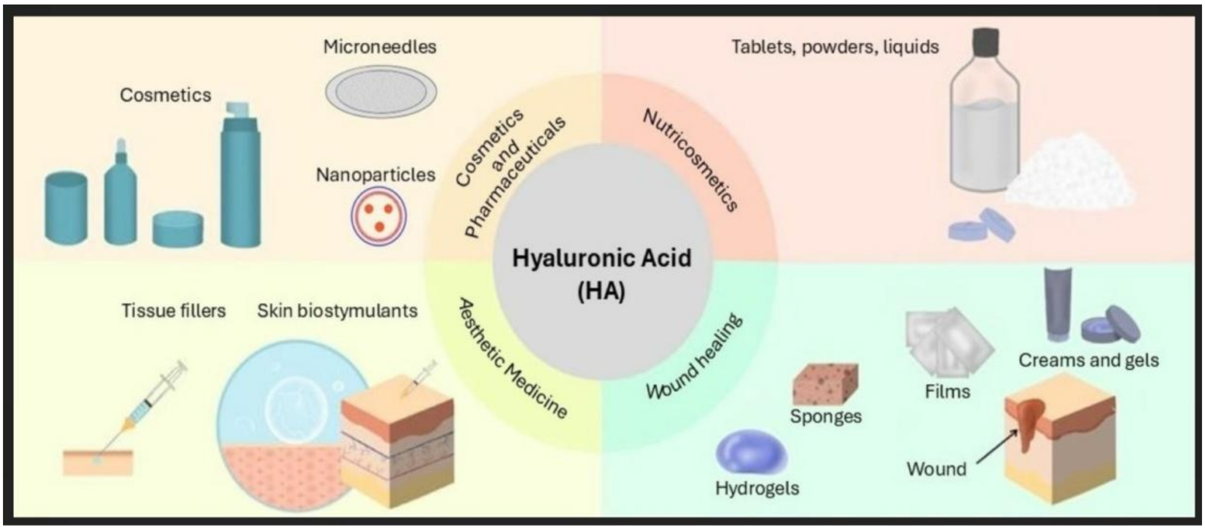
Within the dermis, a key feature is the extracellular matrix (ECM), which is like a gel-filled network surrounding the skin cells. The ECM contains proteins such as collagen and elastin, which provide structural support and elasticity. Collagen is responsible for strength, while elastin allows the skin to stretch and return to its original shape. Another important component of the ECM is glycosaminoglycans (GAGs), which are sugar-based molecules that attract and hold water. Among these, hyaluronic acid is the most abundant and the most important for hydration. Because of its hydrophilic (water-loving) nature, HA can bind up to 1,000 times its weight in water, making it a natural reservoir that keeps the skin moist and supple[5].

The natural role of HA in the skin goes beyond just holding water. It also helps maintain skin turgor, which is the firmness and plumpness that comes from healthy hydration. By creating a hydrated gel-like environment in the ECM, HA supports the stability of collagen and elastin fibers, preventing them from collapsing. This balance allows the skin to stay smooth, elastic, and resistant to mechanical stress. HA also plays a role in signaling between cells in the skin, helping them respond to injury and regulate repair processes[6].

However, the levels of HA in the skin are not constant throughout life. One of the key changes that comes with aging is the decline of HA. This happens for two main reasons. First, the body naturally produces less HA with age because of reduced activity of the enzymes that make it (called hyaluronan synthases). Second, the existing HA is broken down faster by enzymes called hyaluronidases, which become more active as the skin ages. As a result, older skin has less HA available to retain water and maintain ECM stability. The outcome is visible dryness, fine lines, and a loss of elasticity. Wrinkles and sagging are not only caused by the breakdown of collagen and elastin but also by the reduction of HA that normally supports them .

[5] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. Dermato-Endocrinology, 4(3), 253–258. https://doi.org/10.4161/derm.21923

[6] (Shang, L., Li, M., Xu, A., & Zhuo, F. (2024). Recent applications and molecular mechanisms of hyaluronic acid in skin aging and wound healing. *Medicine in Novel Technology and Devices*, *22,* 100320. https://doi.org/10.1016/j.medntd.2024.100320



Chylińska, N., & Maciejczyk, M. (2025). Hyaluronic acid and skin: Its role in aging and wound-healing processes. Gels, 11(4), 281. *https://doi.org/10.3390/gels11040281*

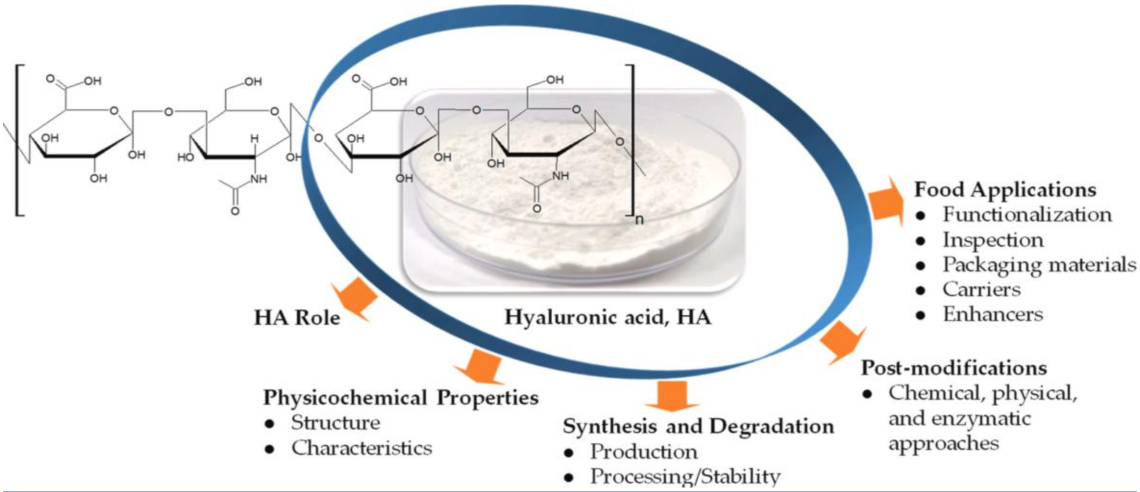
Chemistry and Biology of Hyaluronic Acid

Hyaluronic acid (HA) is a linear polysaccharide made of repeating pairs (disaccharides) of two simple sugars: D-glucuronic acid and N-acetyl-D-glucosamine. These pairs are linked in a long chain that attracts and holds water because the molecule is highly hydrophilic (water-loving). In skin, HA sits outside cells in the extracellular matrix (ECM) and acts like a soft, hydrated gel that supports cells and allows nutrients and signals to move through the tissue. This simple chemical design—repeating sugar units with many sites that bind water—explains why HA can trap large volumes of water and why it is central to skin moisture and flexibility[7].

A key feature of HA biology is molecular weight (size). Natural HA in healthy tissue is usually high-molecular-weight HA (HMW-HA)—often above ~1,000 kDa. HMW-HA forms a quiet,

[7] Dicker, K. T., Gurski, L. A., Pradhan-Bhatt, S., Witt, R. L., Farach-Carson, M. C., & Jia, X. (2014). Hyaluronan: A simple polysaccharide with diverse biological functions. Acta Biomaterialia, 10(4), 1558–1570. https://doi.org/10.1016/j.actbio.2013.12.019).

protective background in the ECM: it stabilizes the gel, reduces friction between fibers, and tends to calm inflammation. When HA chains are cut into low-molecular-weight pieces (LMW-HA)—tens to a few hundreds of kDa—or even shorter oligosaccharides, the smaller fragments behave differently. They move more easily through tissue and can send danger or repair signals to cells. In general terms, HMW-HA is associated with tissue integrity and anti-inflammatory tone, while LMW-HA fragments can activate immune and repair pathways, a size-dependent "language" that the skin uses during stress and healing. This "size matters" principle is well documented in reviews of HA biology and skin aging[8].



Source - Cheng, Q., Liu, C., Zhao, J., Li, W., Guo, F., Qin, J., & Wang, Y. (2023). Unlocking the potential of hyaluronic acid: Exploring its physicochemical properties, modification, and role in food applications. Trends in Food Science & Technology, 139, 104218. https://doi.org/10.1016/j.tifs.2023.104218

Cells "sense" HA through receptors on their surface. The best known is CD44, a transmembrane protein found on keratinocytes, fibroblasts, endothelial cells, and many immune

[8] Bravo, B., Correia, P., et al. (2022). Benefits of topical hyaluronic acid for skin quality and signs of skin aging: From literature review to clinical evidence. Dermatologic Therapy, 35(12), e15903. https://doi.org/10.1111/dth.15903

cells. When HA binds CD44, it can change cell shape, movement, and gene activity—affecting barrier function, hydration behavior, and matrix production. Another HA-binding protein is RHAMM (Receptor for Hyaluronan-Mediated Motility, also called CD168), which participates in cell migration and cytoskeletal changes, especially during repair. In addition, smaller HA fragments can engage Toll-like receptors (TLR2 and TLR4), which are pattern-sensing proteins typically used by the immune system to detect damage. This receptor network helps explain why HA can do more than just hold water: it can signal fibroblasts to make collagen, guide cells during wound closure, and tune local inflammation up or down depending on HA size[9].

Size also shapes which receptors talk to each other. For example, studies show that LMW-HA can promote CD44–TLR complexes, amplifying downstream signals that drive defense and remodeling—useful in healing, but potentially pro-inflammatory if not controlled. By contrast, HMW-HA supports tissue homeostasis, helps maintain a hydrated ECM, and tends to dampen TLR signaling, which fits with its protective role in healthy skin. In practical terms, this means that a formulation rich in HMW-HA will primarily support moisture and calm, while specific LMW-HA fractions may be chosen to nudge repair pathways—a concept increasingly explored in dermatology[10].

HA is made at the cell membrane by enzymes called hyaluronan synthases (HAS1, HAS2, HAS3). These enzymes pull sugar units from the cell's internal pool and extrude the HA chain directly outside the cell, where it immediately joins the ECM. The three synthases differ in how fast

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[9] Shatirishvili, M., Burk, A. S., *et al.* (2016). Epidermal-specific deletion of CD44 reveals a function in keratinocytes in response to mechanical stress. *Cell Death & Disease, 7*(12), e2461. https://doi.org/10.1038/cddis.2016.360)

[10] (Misra, S., Hascall, V. C., Markwald, R. R., & Ghatak, S. (2015). Interactions between hyaluronan and its receptors (CD44, RHAMM) regulate the activities of inflammation and cancer. Frontiers in Immunology, 6, 201. https://doi.org/10.3389/fimmu.2015.00201)

they work and the chain lengths they tend to produce, which lets skin cells tailor HA output to local needs (for example, producing longer chains for stability or shorter chains during repair). Changes in HAS expression are seen in skin aging and in wound healing models, supporting the idea that synthesis control is part of normal skin adaptation[11].

HA in skin also turns over quickly. It is degraded by enzymes called hyaluronidases (such as HYAL1 and HYAL2) and by other proteins that process HA at the cell surface and within tissues. The half-life of HA is short—measured in hours in the epidermis and roughly a day (or so) in the dermis—which means skin is constantly making and breaking HA to match daily needs and respond to stress. This rapid turnover helps skin stay adaptable, but it also explains why cross-linking or delivery technologies are used in treatments to extend HA's presence and effects. Scientifically, knowing the half-life and the degradative pathways helps clinicians choose molecular weights and formulations that fit a goal (surface hydration vs. deeper, longer-lasting support).[12]

[11] Wang, Y., Lauer, M. E., et al. (2014). Hyaluronan synthase 2 protects skin fibroblasts against apoptosis induced by environmental stress. The Journal of Biological Chemistry, 289(46), 32253–32265. https://doi.org/10.1074/jbc.m114.578377)

[12] Wang, Y., Lauer, M. E., et al. (2014). Hyaluronan synthase 2 protects skin fibroblasts against apoptosis induced by environmental stress. The Journal of Biological Chemistry, 289(46), 32253–32265. https://doi.org/10.1074/jbc.m114.578377)

Figure 2. The role of hyaluronic-acid-based hydrogel in wound healing.

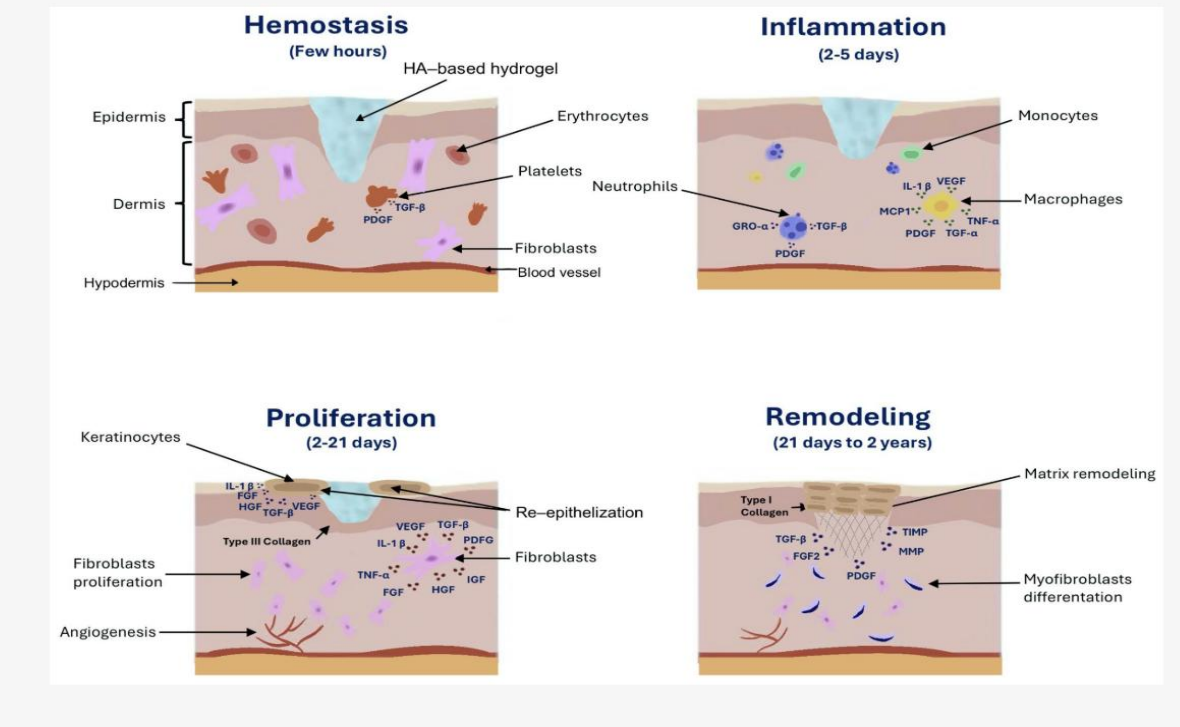

Figure . The role of hyaluronic-acid-based hydrogel in wound healing.

Chylińska, N., & Maciejczyk, M. (2025). Hyaluronic acid and skin: Its role in aging and wound-healing processes. Gels, 11(4), 281. https://doi.org/10.3390/gels11040281

Mechanisms of Skin Hydration through Hyaluronic Acid

One of the most important reasons hyaluronic acid (HA) is widely used in dermatology and skincare is its ability to improve skin hydration. Hydration is essential for keeping the skin soft, plump, and healthy, and HA influences this process at both a physical and cellular level.

The first and most direct mechanism is HA's water-binding property. Each HA molecule can hold up to 1,000 times its own weight in water, which makes it one of the most powerful natural moisturizers in the body.[13]In the skin, HA molecules are found in the extracellular matrix (ECM), the gel-like substance surrounding cells in the dermis and epidermis. By pulling and storing water in this

[13] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923

space, HA creates a reservoir of moisture that keeps tissues well hydrated. This ability not only plumps the skin but also improves turgor (the firmness of the skin when pinched or pressed), which is a direct sign of hydration.

Another key mechanism is HA's role in enhancing the barrier function of the skin. The stratum corneum, the outermost layer of the epidermis, is the body's first defense against dehydration. When this barrier becomes weak, water escapes from the skin surface in a process called transepidermal water loss (TEWL). HA helps reduce TEWL by maintaining moisture in the upper skin layers and reinforcing the natural lipid-protein barrier. Clinical studies show that topical HA formulations improve hydration in the stratum corneum and reduce dryness, making the skin feel smoother and less flaky[14]. By keeping the barrier intact, HA protects against external stressors such as pollution, UV radiation, and harsh climates, all of which can accelerate skin aging.

Beyond its physical water-holding role, HA also influences hydration through cell signaling pathways. One example is its interaction with aquaporins, a family of proteins that form water channels in cell membranes. In the skin, Aquaporin-3 (AQP3) is especially important because it transports both water and glycerol, two molecules essential for skin hydration and elasticity. Research has shown that HA treatments can stimulate the expression of AQP3 in keratinocytes (the main cells of the epidermis). This means that HA does not just trap water in the extracellular space but also helps regulate how water moves between cells, improving the distribution of moisture within

[14] Osseiran, S., Dela Cruz, J., *et al.* (2019). Characterizing stratum corneum structure, barrier function, and chemical content of human skin with coherent Raman scattering imaging. *Biomedical Optics Express*, *10*(5), 2542–2555. https://doi.org/10.1364/BOE.10.002542)

the skin layers[15]. This dual action—external water storage plus improved internal water transport—makes HA a unique tool for long-lasting hydration.

HA also interacts with fibroblasts and keratinocytes, the two most important cell types in skin hydration. Fibroblasts, found in the dermis, are responsible for producing collagen, elastin, and glycosaminoglycans such as HA itself. When HA binds to receptors like CD44 on fibroblasts, it can activate signaling pathways that increase HA production and improve ECM hydration (Frontiers). Keratinocytes, located in the epidermis, respond to HA by improving cell differentiation and renewal, which strengthens the barrier and promotes an even distribution of water throughout the epidermis. Together, these cell-level effects ensure that hydration is not only maintained at the surface but also supported at deeper levels of the skin.[16]

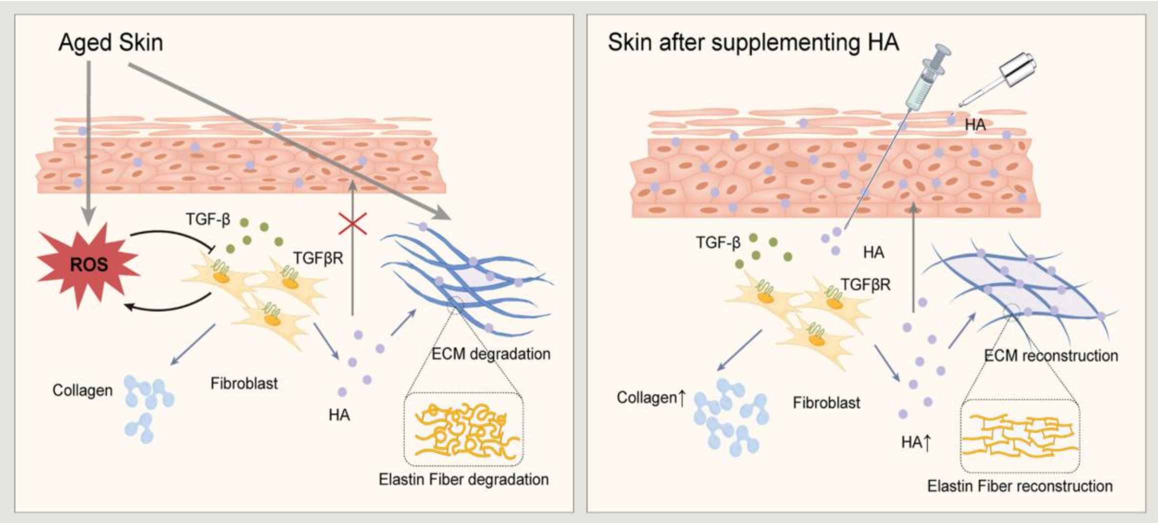

Fig. Illustrates the anti-aging process contributed by the application of HA-based formulations. This figure generated by Figdraw (https://www.figdraw.com/).

[15] (Karimi, N., & Ahmadi, V. (2024). Aquaporin channels in skin physiology and aging pathophysiology: Investigating their role in skin function and the hallmarks of aging. *Cells, 13*(9), 784. https://doi.org/10.3390/cells13090784).

[16] (Bravo, B., Correia, P., et al. (2022). Benefits of topical hyaluronic acid for skin quality and signs of skin aging: From literature review to clinical evidence. Dermatologic Therapy, 35(12), e15903. https://doi.org/10.1111/dth.15903)

Source - Shang, L., Li, M., Xu, A., & Zhuo, F. (2024). Recent applications and molecular mechanisms of hyaluronic acid in skin aging and wound healing. Medicine in Novel Technology and Devices, 22, 100320. https://doi.org/10.1016/j.medntd.2024.100320

Finally, it is important to note that hydration provided by HA has a visible impact on skin appearance. Well-hydrated skin reflects light better, feels softer, and shows fewer fine lines and wrinkles. This explains why HA-based serums and dermal fillers are so effective in improving the look of dry or aged skin. Clinical research confirms that HA treatments—whether topical, injectable, or advanced delivery forms—lead to measurable improvements in hydration and skin smoothness, with results often seen within weeks.[17]

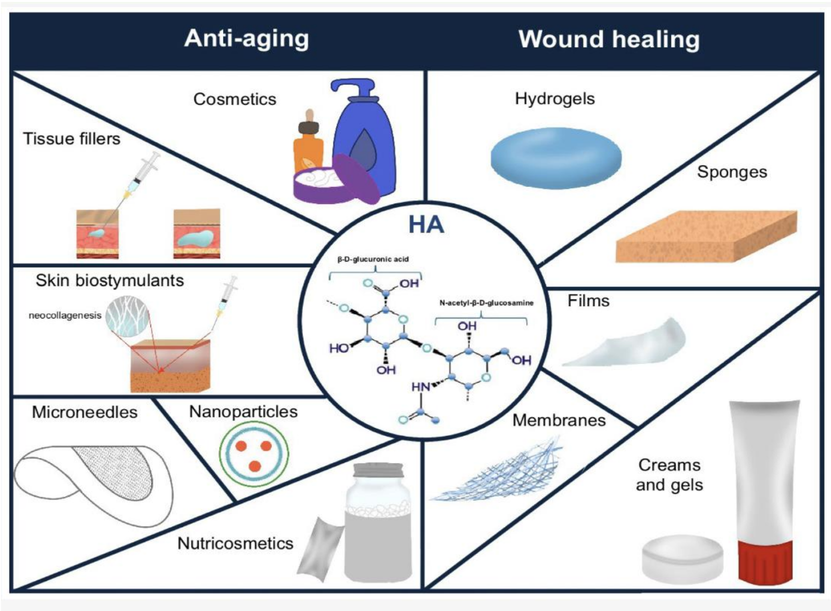

Figure . The use of hyaluronic acid (HA) in aging prevention and wound healing.

Source - Chylińska, N., & Maciejczyk, M. (2025). Hyaluronic acid and skin: Its role in aging and wound-healing processes. Gels, 11(4), 281. https://doi.org/10.3390/gels11040281

[17] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923).

Mechanisms of Skin Elasticity through Hyaluronic Acid

Skin elasticity depends on how well the extracellular matrix (ECM) holds together and moves. Collagen gives strength, elastin gives stretch, and hyaluronic acid (HA) creates the hydrated "gel" that spaces these fibers so they can glide without sticking. By drawing and holding water in this space, HA keeps the matrix soft and flexible, supports fiber alignment, and helps the dermis resist daily bending and compression—key reasons why HA-rich skin feels springy rather than stiff. Reviews of skin biology and aging describe HA as a central ECM component that preserves the viscoelastic behavior of collagen–elastin networks[18].

Beyond this physical support, HA also signals skin cells to build a better matrix. When HA binds to CD44 on dermal fibroblasts (the cells that make collagen and elastin), it switches on intracellular pathways such as MAPK/ERK and PI3K–Akt. These signals encourage fibroblasts to spread, move, and make more type I collagen and components that stabilize elastic fibers. In simple terms, HA not only cushions the matrix; it "talks" to the cells that maintain that matrix, nudging them toward repair and renewal. Foundational papers and recent summaries of CD44 biology show this HA–CD44–MAPK/ERK link and its downstream effects on matrix genes[19]. [20]

Cross-linked HA dermal fillers add a second, complementary route to elasticity: mechanical support and mechanotransduction. Cross-linking turns HA into a longer-lasting, elastic gel that

[18] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology, 4*(3), 253–258. https://doi.org/10.4161/derm.21923)

[19] (Meran, S., Luo, D. D., et al. (2011). Hyaluronan facilitates transforming growth factor-β1-dependent proliferation via CD44 and epidermal growth factor receptor interaction. The Journal of Biological Chemistry, 286(20), 17618–17630. https://doi.org/10.1074/jbc.M110.195701)

[20] Yue, Z., Shao, K. (2025). Visualization of the relationship between hyaluronic acid and wound healing: A bibliometric analysis. Skin Research and Technology, 31(5), e70164. https://doi.org/10.1111/srt.70164)

immediately props up relaxed or thinned dermis. Fibroblasts sense this gentle stretch, reorganize their cytoskeleton, and begin new collagen production ("neocollagenesis"). Human and ex vivo studies show that injecting cross-linked HA stimulates type I collagen in photodamaged or elderly skin, with histology confirming new fibers after treatment. These effects blend the mechanical (space filling, tissue support) and the biologic (cell signaling) to gradually improve firmness beyond the initial volume effect[21].

HA can also amplify growth-factor pathways that drive matrix rebuilding, notably TGF-β/Smad. Experimental work indicates that HA-based fillers (and some HA composites) can enhance TGF-β signaling in dermal tissue, which in turn upregulates genes for collagen and elastic fiber assembly. This is consistent with clinical observations that elasticity improves over weeks to months, a timeline that aligns with biologic remodeling rather than simple water retention alone[22].

Finally, HA helps protect the matrix from oxidative stress. Reactive oxygen species (ROS) break down collagen and elastin and cross-link proteins in ways that stiffen the dermis. Laboratory and review data show that HA—especially low-molecular-weight HA in some models—can scavenge free radicals and modulate ROS-related signaling, reducing damage to ECM components. In practice, HA in fillers is often paired with antioxidants such as mannitol to further limit oxidative stress during and after injection, supporting a healthier environment for new collagen and elastin to form. These antioxidant and ROS-modulating effects add another layer to HA's pro-elasticity profile.

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[21] (Wang, F., Do, T. T., *et al.* (2024). Implications for cumulative and prolonged clinical improvement induced by cross-linked hyaluronic acid: An in vivo biochemical/microscopic study in humans. *Experimental Dermatology, 33*(2), 14998. https://doi.org/10.1111/exd.14998)

[22] (Fan, Y., Tae-Hyun, C., *et al.* (2019). Hyaluronic acid-cross-linked filler stimulates collagen type 1 and elastic fiber synthesis in skin through the TGF-β/Smad signaling pathway in a nude mouse model. *Journal of Plastic, Reconstructive & Aesthetic Surgery, 72*(7), 1154–1163. https://doi.org/10.1016/j.bjps.2019.03.032)

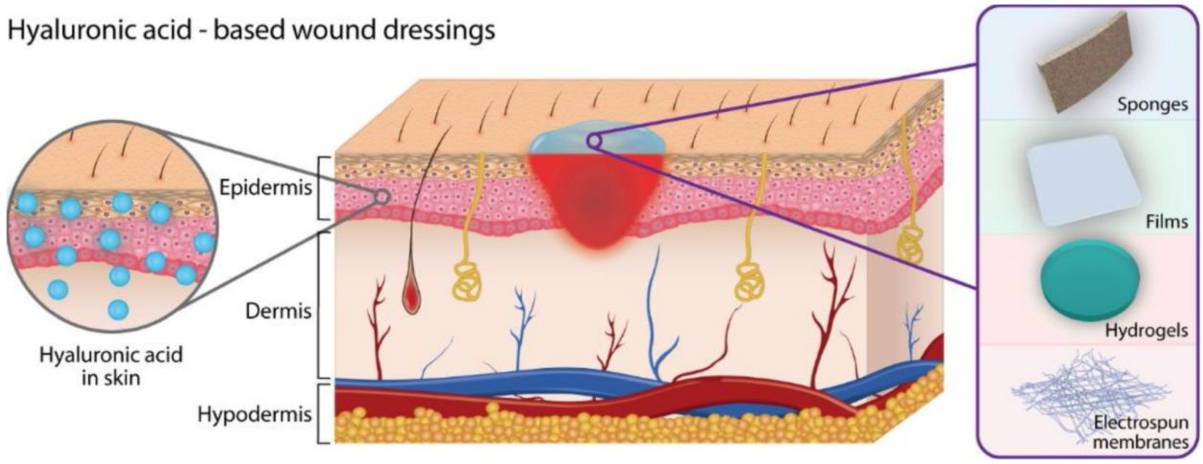

Figure - Hyaluronic acid (HA), based wound dressings

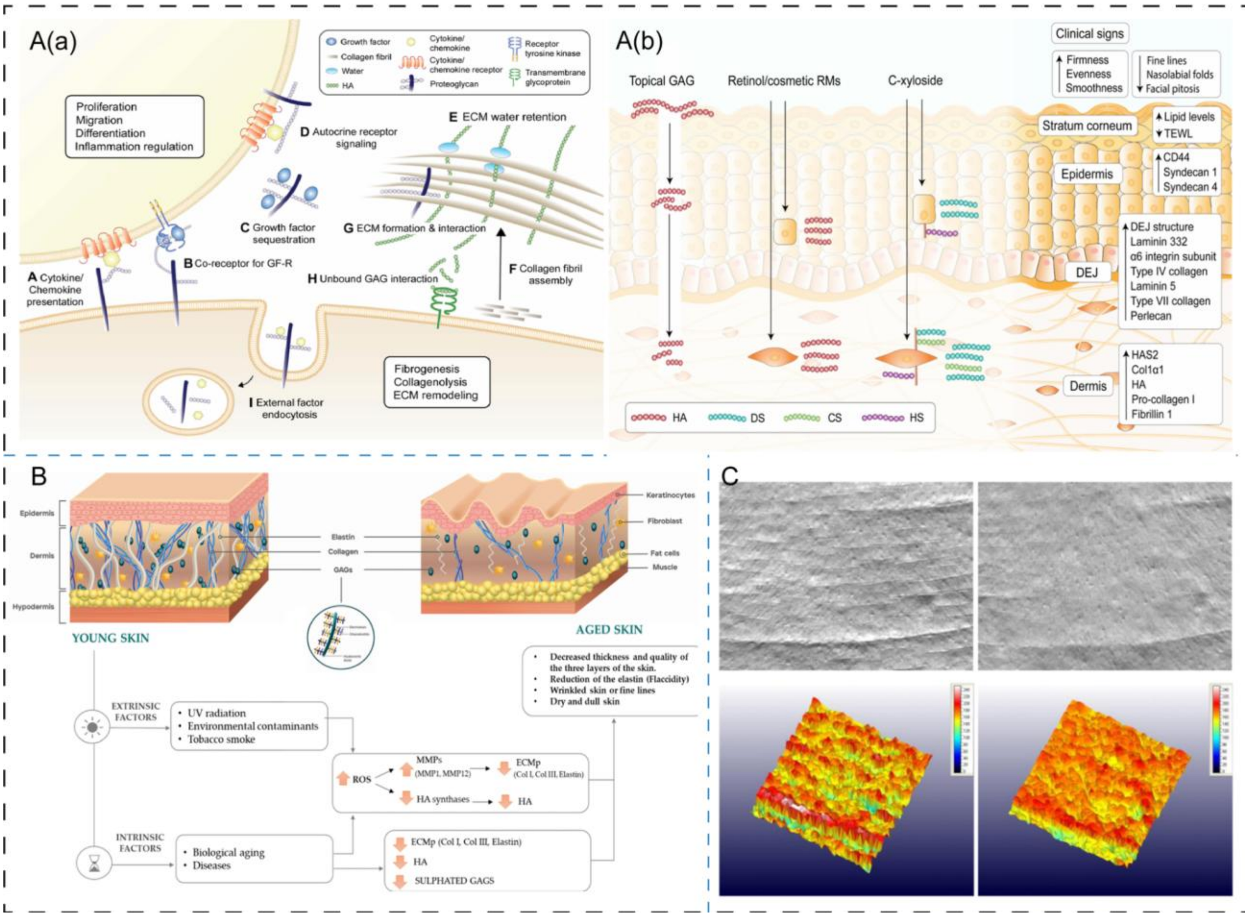

Fig. HA and other components in skin aging. (A) Key biological activities of GAG & proteoglycans in skin (a); Main strategies of exploiting GAGs for skincare (b). (B) Changes in skin structure and composition due to the aging process. (C) The images of skin texture before product application and after 28 days of regular application of an anti-ageing cream incorporating 0.5 % LMW-HA (20–50 KDa) and 3 % HMW-HA (1–1.4 MDa). Reproduced and adapted with permission: (A) [63] Copyright 2021, Dove. (B) [64] Copyright 2023, MDPI. (C) [66] Copyright 2021, MDPI.

Source -Shang, L., Li, M., Xu, A., & Zhuo, F. (2024). Recent applications and molecular mechanisms of hyaluronic acid in skin aging and wound healing. Medicine in Novel Technology and Devices, 22, 100320. https://doi.org/10.1016/j.medntd.2024.100320

Clinical Applications of HA-Based Treatments

The unique chemistry and biology of hyaluronic acid (HA) have made it one of the most widely used molecules in modern dermatology and cosmetic medicine. Its ability to hold water, interact with skin cells, and remodel the extracellular matrix (ECM) explains why HA-based products are applied in many different forms, from simple creams to injectable fillers. Each form of treatment has its own benefits, limitations, and underlying cellular effects.

One of the most common uses is topical HA, usually found in serums, gels, and creams. These products focus on boosting surface hydration and improving the skin barrier. Because HA molecules are large, they cannot easily penetrate very deep into the skin. High-molecular-weight HA mostly stays on the surface and improves the stratum corneum (outer layer) by reducing water loss and giving a smoother feel. Low-molecular-weight HA, on the other hand, can move deeper into the epidermis and influence cell behavior by signaling keratinocytes and fibroblasts[23] . Clinical studies

[23] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923).

show that topical HA products improve skin hydration, reduce roughness, and soften fine lines, although the effects are temporary and require regular application[24].

For longer-lasting and deeper effects, injectable HA is widely used in the form of dermal fillers. These products are made of cross-linked HA, which means the molecules are chemically connected to resist quick breakdown. When injected into the dermis, fillers restore lost volume, smooth out wrinkles, and improve skin elasticity. More importantly, research shows that injectable HA does more than just fill space: it also stimulates fibroblasts, leading to the production of new collagen and elastin, a process called neocollagenesis (Frontiers). This means the benefits extend beyond the initial plumping effect, improving the skin's structure over months. Side effects are generally mild, such as redness or swelling, but proper technique is critical to avoid rare complications[25].

Recent research has focused on improving how HA is delivered to the skin. Nanoparticle and liposomal HA formulations are being developed to overcome the penetration limits of standard HA. By packing HA into microscopic carriers, these systems can deliver it deeper into the skin layers, protect it from quick breakdown, and release it more slowly over time. Early clinical studies suggest that nano-HA and liposomal HA provide stronger and longer-lasting hydration compared to traditional creams (PMC). Another emerging method is microneedle patches, which create tiny, painless channels in the skin to deliver HA directly into the dermis, mimicking some effects of injectable treatments without the need for needles.

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[24] (Juncan, A. M., Moisă, D. G., Santini, A., Morgovan, C., Rus, L.-L., Vonica-Țincu, A. L., & Loghin, F. (2021). Advantages of hyaluronic acid and its combination with other bioactive ingredients in cosmeceuticals. *Plants, 10*(5), 941. https://doi.org/10.3390/plants10050941).

[25] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923).

Finally, HA is also being combined with other active ingredients for synergistic effects. For example, HA with peptides can support both hydration and collagen production, HA with retinoids can balance moisture while improving cell turnover, and HA with vitamin C can combine hydration with antioxidant protection[26] . These combination therapies show how HA can act as a "base" molecule that enhances the performance of other treatments while keeping the skin barrier intact.

Cellular Mechanisms of HA in Anti-Aging

Hyaluronic acid (HA) helps skin look and act younger not only by holding water, but also by guiding how skin cells behave. A major anti-aging action is inflammation control. In healthy tissue, high-molecular-weight HA (HMW-HA) tends to calm immune signaling, while low-molecular-weight fragments (LMW-HA)—often created during stress or injury—can trigger alarms through Toll-like receptors (TLR2/4) and downstream NF-κB/MAPK pathways. In simple terms, intact HA supports a quiet, balanced environment, but fragments can call in short-term help for repair. This size-dependent effect explains why many treatments favor HMW-HA to help limit pro-inflammatory cytokines such as IL-1 and TNF-α, supporting a smoother, less reactive skin state[27].

HA also supports cell proliferation and migration, two steps that matter for everyday maintenance and wound repair. By binding CD44 on keratinocytes (epidermal cells) and fibroblasts (dermal matrix-building cells), HA encourages cells to move into damaged areas and divide, helping close micro-injuries that accumulate with age. Experiments using human skin cells show that adding

[26] (Juncan, A. M., Moisă, D. G., Santini, A., Morgovan, C., Rus, L.-L., Vonica-Țincu, A. L., & Loghin, F. (2021). Advantages of hyaluronic acid and its combination with other bioactive ingredients in cosmeceuticals. *Plants, 10*(5), 941. https://doi.org/10.3390/plants10050941

[27] (Ruppert, S. M., Hawn, T. R., et al. (2014). Tissue integrity signals communicated by high-molecular weight hyaluronan and the resolution of inflammation. Frontiers in Immunology, 5, 389. https://doi.org/10.3389/fimmu.2014.00389) .

HA—especially in the right size range—boosts these repair behaviors, which over time helps preserve a fresher surface and stronger barrier[28].

A third pathway is angiogenesis—the formation of tiny blood vessels that feed the skin. Endothelial cells (the cells that line vessels) use CD44 to sense HA; appropriate HA cues promote endothelial migration, tube formation, and vessel growth. Better microcirculation means better delivery of oxygen and nutrients and more efficient waste removal, all of which support youthful tissue turnover and resilience[29]. At the gene level, HA signaling can shift the balance toward building rather than breaking the matrix. Cross-linked HA fillers provide gentle mechanical support and, via mechanotransduction and CD44-linked pathways (e.g., MAPK/ERK), stimulate fibroblasts to make type I/III collagen and elastic-fiber components. Classic human studies in photo-damaged skin show increased procollagen markers weeks after HA injection—evidence of true neocollagenesis, not just water plumping. Some HA-based formulations (and HA combined with growth factors) also reduce matrix metalloproteinases (MMP-1/3/9), the enzymes that break down collagen, further tilting the system toward repair. Together, upregulating collagen/elastin and downregulating MMPs helps skin regain firmness and spring.[30]

Comparative Efficacy of HA Treatments

The benefits of hyaluronic acid (HA) depend on molecular size and how it is delivered to the skin. In general, high–molecular weight HA (HMW-HA) forms a calm, hydrated gel in the

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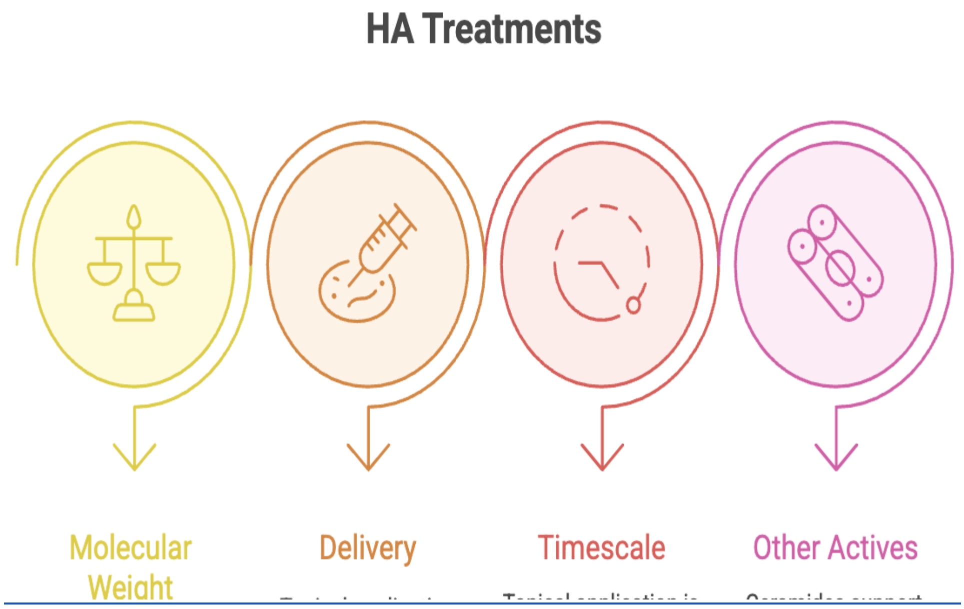
[28] (Frenkel, J. S. (2014). The role of hyaluronan in wound healing. The Journal of Craniofacial Surgery, 25(1), 130–133. https://doi.org/10.1097/SCS.0000000000000300) .

[29] (Frenkel, J. S. (2014). The role of hyaluronan in wound healing. The Journal of Craniofacial Surgery, 25(1), 130–133. https://doi.org/10.1097/SCS.0000000000000300)

[30] Wang, F., Do, T. T., et al. (2024). Implications for cumulative and prolonged clinical improvement induced by cross-linked hyaluronic acid: An in vivo biochemical/microscopic study in humans. Experimental Dermatology, 33(2), 14998. https://doi.org/10.1111/exd.14998) .

extracellular matrix and tends to reduce inflammation while improving surface moisture; low–molecular weight HA (LMW-HA) and HA oligosaccharides penetrate more easily and can activate repair signals in cells through receptors such as CD44 and Toll-like receptors (TLR2/4). This “size code” explains why HMW-HA excels at soothing hydration, whereas select LMW fractions are often used to nudge remodeling pathways that support.[31]



Topical HA (serums/creams) mainly improves short-term hydration and feel. Because larger HA stays near the surface, it reduces water loss and smooths roughness; smaller HA can reach the upper epidermis and influence keratinocyte behavior. Randomized clinical trials show measurable gains in hydration and skin quality with topical HA, though effects require regular use to maintain[32]. By contrast, injectable, cross-linked HA (dermal fillers) delivers both immediate mechanical support and longer-term biological change. Beyond filling space, well-designed studies in photo-damaged

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[31] elasticity (Hu, L., Nomura, S., et al. (2022). Anti-inflammatory effects of differential molecular weight hyaluronic acids on UVB-induced calprotectin-mediated keratinocyte inflammation. Experimental Gerontology, 165, 111851. https://doi.org/10.1016/j.exger.2022.111851)

[32] Bravo, B., Correia, P., *et al.* (2022). Benefits of topical hyaluronic acid for skin quality and signs of skin aging: From literature review to clinical evidence. *Dermatologic Therapy*, *35*(12), e15903. https://doi.org/10.1111/dth.15903)

human skin have shown neocollagenesis—new type I collagen—weeks to months after injection, consistent with CD44/MAPK signaling and mechanotransduction in dermal fibroblasts; this underlies durable gains in firmness and elasticity that outlast simple water binding[33].

A third route is oral HA supplements, which aim for body-wide support of skin moisture. Controlled trials report increased skin hydration within 2–8 weeks (sometimes with improved tone or epidermal thickness after longer use), suggesting that absorbed HA or its metabolites can influence water balance and matrix turnover from within. While dose, source, and product quality vary, the best evidence indicates a modest but real hydration benefit in both younger and older adults[34].

When comparing time scales, topical HA offers fast, reversible improvements in the stratum corneum; injectables add structural and cellular effects that build over weeks to months; and oral HA provides systemic, gradual hydration support that depends on continuous intake. These differences reflect distinct cellular mechanisms: surface TEWL reduction (topical), fibroblast activation and ECM remodeling (injectable), and potential modulation of cutaneous hydration pathways via circulating metabolites (oral)[35].

It also helps to compare HA with other common actives. Ceramides (the skin's natural barrier lipids) strengthen the outer layer and can improve dryness and barrier scores in conditions like atopic dermatitis, though meta-analyses suggest they are not always superior to other moisturizers for

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[33] (Wang, F., Garza, L. A., *et al.* (2007). In vivo stimulation of de novo collagen production caused by cross-linked hyaluronic acid dermal filler injections in photodamaged human skin. *JAMA Dermatology*, *143*(2), 155–163. https://doi.org/10.1001/archderm.143.2.155)

[34] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology, 4*(3), 253–258. https://doi.org/10.4161/derm.21923)

[35] (Wang, F., Garza, L. A., et al. (2007). In vivo stimulation of de novo collagen production caused by cross-linked hyaluronic acid dermal filler injections in photodamaged human skin. JAMA Dermatology, 143(2), 155–163. https://doi.org/10.1001/archderm.143.2.155)

reducing TEWL in every setting; they are best viewed as barrier builders, complementary to HA's water-holding and signaling roles[36]. Collagen peptides taken orally have growing high-quality evidence: recent systematic reviews and meta-analyses across multiple RCTs show significant improvements in hydration and elasticity, likely by supplying amino acids and signaling peptides that stimulate dermal collagen synthesis. These data position oral collagen as a useful adjunct to HA—collagen focuses on fiber building, while HA optimizes the hydrated matrix that surrounds those fibers[37].

Limitations and Challenges

Hyaluronic acid (HA) treatments are effective, but they also face clear limits in the real world of skin biology and clinical use. The first challenge is rapid degradation of HA in skin. Native HA turns over quickly: studies report a half-life of only a few hours in the epidermis and roughly one day in the dermis, where it is continuously broken down by hyaluronidases and cleared from tissue. This fast metabolism explains why simple hydration effects fade and why formulations try to slow breakdown (for example, by cross-linking HA in fillers)[38].

Because HA is cleared so quickly, many patients need repeated treatments. Topical HA requires regular, ongoing use. Injectable, cross-linked HA lasts longer, but longevity still varies by

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[36] (Kono, T., Miyachi, Y., et al. (2022). Clinical significance of the water retention and barrier function-improving capabilities of ceramide-containing formulations: A qualitative review. The Journal of Dermatology, 49(8), 767–777. https://doi.org/10.1111/1346-8138.16401)

[37] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923)

[38] .(Buhren, B. A., Schrumpf, H., et al. (2016). Hyaluronidase: From clinical applications to molecular and cellular mechanisms. European Journal of Medical Research, 21, 5. https://doi.org/10.1186/s40001-016-0201-5)

product, placement, and patient—prospective and review data show durability from many months to beyond a year, sometimes longer, yet retreatment is common to maintain results[39].

A second challenge is variability in patient response. Skin thickness, baseline hydration, sun damage, age, and even local biomechanics change how HA performs. On top of that, product differences (e.g., HA concentration, cross-linking density, cohesivity, elasticity) and technique (depth, volume, injection pattern) strongly influence outcomes. Comparative reviews emphasize that fillers with different rheology behave differently in tissue, which partly explains why two patients (or two facial areas) can respond in different ways to the same “HA” label. This heterogeneity also makes it hard to compare studies or predict exact longevity for a given person[40].

Safety is generally good, but side effects matter. Common reactions include swelling, redness, tenderness, and bruising that settle within days. Less common events include delayed

nodules, granulomatous reactions, and biofilm-related inflammation, which may appear weeks to months later and can require targeted management. Rare but serious complications—notably vascular occlusion leading to skin necrosis or even vision loss—have been documented, underscoring the importance of anatomy-aware technique and rapid response protocols. Large reviews catalog these events and stress prevention (appropriate cannula/needle choice, low injection pressure, careful placement) and treatment readiness (e.g., hyaluronidase to dissolve HA in urgent scenarios)[41].

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[39] (Siperstein, R., Nestor, E., et al. (2023). One-year data on the longevity and safety of hyaluronic acid filler for static horizontal neck rhytids. Dermatologic Surgery, 49(11), 1153–1160. https://doi.org/10.1097/DSS.0000000000003896)

[40] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923

[41] (Buhren, B. A., Schrumpf, H., et al. (2016). Hyaluronidase: From clinical applications to molecular and cellular mechanisms. European Journal of Medical Research, 21, 5. https://doi.org/10.1186/s40001-016-0201-5

There are also limits in evidence and access. For topical HA, penetration into deeper layers is restricted by molecular size; while low-molecular-weight or nano-HA can improve penetration, results vary and many studies are small or short-term. This creates uncertainty about how long benefits last without continuous use[42]. For injectables, differences in outcome measures and follow-up windows across trials make true cost-effectiveness comparisons difficult; repeated sessions and practitioner expertise add to overall cost and accessibility concerns, particularly outside urban centers. Methodological critiques call for more standardized testing so clinicians and patients can compare products and plan treatment intervals more reliably[43].

Future Directions

Future work on hyaluronic acid (HA) will likely focus on making treatments last longer, reach the right skin layer, and do more than hydration—all while staying safe. First, bioengineered HA will keep improving. New cross-linked and hybrid gels (for example, HA blended with collagen or PEG-modified HA) aim to resist fast breakdown, provide better "spring," and trigger steady remodeling rather than a short plumping effect[44].[45] Second, nanotechnology-based delivery is moving fast. HA-modified liposomes and nanoparticles can carry actives (or HA itself) deeper and release them slowly, which could turn once-a-day serums into longer-acting treatments; dissolving

[42] ((Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. Dermato-Endocrinology, 4(3), 253–258. https://doi.org/10.4161/derm.21923

[43] Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology, 4*(3), 253–258. https://doi.org/10.4161/derm.21923

[44] Kim, Z.-H., Lee, Y., et al. (2015). A composite dermal filler comprising cross-linked hyaluronic acid and human collagen for tissue reconstruction. Journal of Microbiology and Biotechnology, 25(3), 399–406. https://doi.org/10.4014/jmb.1411.11029

[45] Buhren, B. A., Schrumpf, H., et al. (2016). Hyaluronidase: From clinical applications to molecular and cellular mechanisms. European Journal of Medical Research, 21, 5. https://doi.org/10.1186/s40001-016-0201-5

HA microneedle patches offer painless, at-home delivery to the upper dermis with growing evidence for photoaging care[46].

Third, pairing HA with regenerative medicine is promising. HA hydrogels are biocompatible "scaffolds" that can host stem cells (for example, mesenchymal stem cells) or growth factors and guide them to rebuild tissue; high-quality reviews in wound and skin repair suggest HA matrices can improve cell survival, angiogenesis, and controlled release—ideas that could translate to anti-aging facial skin[47].

Fourth, gene-level strategies may eventually let skin make more of its own HA. Basic studies show that HAS2 (hyaluronan synthase-2) is a key enzyme for inducible HA production in fibroblasts, and overexpression raises HA and stress resistance—pointing to a realistic target pathway[48].CRISPR tools already work in skin cells and can knock out or activate genes, so future CRISPRa or vector approaches could up-regulate HAS2 (or tune hyaluronidases) in a local, controlled way once safety is proven[49].

Conclusion

[46] Matalqah, S., Lafi, Z., *et al.* (2024). Hyaluronic acid in nanopharmaceuticals: An overview. *Pharmaceutics, 16*(6), 870. https://doi.org/10.3390/pharmaceutics16060870

[47] (Olteanu, G., Neacșu, S. M., *et al.* (2024). Advancements in regenerative hydrogels in skin wound treatment: A comprehensive review. *Polymers, 16*(7), 1278. https://doi.org/10.3390/polym16071278

[48] (Wang, Y., Lauer, M. E., *et al.* (2014). Hyaluronan synthase 2 protects skin fibroblasts against apoptosis induced by environmental stress. *The Journal of Biological Chemistry, 289*(46), 32253–32265. https://doi.org/10.1074/jbc.M114.578377

[49] Huang, Y., Askew, E. B., *et al.* (2016). CRISPR/Cas9 knockout of HAS2 in rat chondrosarcoma chondrocytes demonstrates the requirement of hyaluronan for aggrecan retention. *Matrix Biology, 56,* 74–94. https://doi.org/10.1016/j.matbio.2016.04.004

This paper shows that hyaluronic acid (HA) improves skin hydration and elasticity through a clear dual mechanism that combines biophysics and cell biology. On the biophysical side, HA is a long sugar chain (a glycosaminoglycan) that is highly hydrophilic, meaning it holds very large amounts of water—up to 1,000 times its weight. In the skin's extracellular matrix (ECM), this creates a soft, hydrated gel that keeps tissues moist, supports fiber spacing, and reduces transepidermal water loss (TEWL). Better surface hydration and lower TEWL strengthen the skin barrier, improve turgor (the springy feel of the skin), and make the surface look smoother and less flaky.[50] HA's role is not only passive water storage: it also helps water move within the epidermis through Aquaporin-3 (AQP3) channels, which transport water and glycerol. Evidence links healthy AQP3 activity with better moisture balance, barrier quality, and cell turnover, explaining why HA-based care often leads to both immediate softness and ongoing comfort (PMC; Annals of Dermatology).

On the biological side, HA acts as a signaling molecule that "talks" to skin cells. It binds to CD44 on keratinocytes and fibroblasts and can influence pathways like MAPK/ERK, which promote fibroblast spreading and activity. Activated fibroblasts make more type I collagen and help organize elastic fibers, which support true elasticity rather than just short-term plumping. ***Size matters:*** high-molecular-weight HA (HMW-HA) helps keep tissues calm and hydrated, while specific low-molecular-weight (LMW) fragments can nudge controlled repair processes. Together, these signals remodel the ECM in a balanced way, improving firmness and resilience over time.

Clinical evidence is consistent with these cellular mechanisms. Topical HA (especially blends of molecular sizes) improves stratum-corneum hydration, reduces roughness, and softens fine lines, with benefits that track with regular use. Injectable, cross-linked HA (dermal fillers) offers both immediate support and longer-term remodeling: beyond filling space, well-designed human studies

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[50] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923).

report neocollagenesis—new collagen formation—after treatment, which aligns with the CD44/ERK and mechanotransduction pathways discussed above. Advanced delivery systems—such as nano-HA, liposomal carriers, and microneedle patches—aim to carry HA deeper, protect it from quick breakdown, and release it steadily, and early studies are promising for sustained hydration and texture gains. Across these modalities, the pattern is the same: HA improves how skin looks because it improves how skin works at the cellular level.[51]

At the same time, real-world outcomes depend on dose, depth, and design. Native HA turns over quickly in skin, so cross-linking (for fillers) and smart formulation (for topicals) are used to extend effects. Patient factors—age, sun damage, barrier status—and product properties—molecular size, cohesivity, elasticity—shape results and explain why treatments must be tailored. Safety is generally good, but expert technique and proper indications matter to lower the risk of adverse events, particularly with injectables. These practical points do not weaken the core science; they remind us that biology + delivery determines the final benefit[52].

Looking ahead, bioengineered HA (hybrids, new cross-links), nanocarriers, and dissolving HA microneedles are likely to make hydration more durable and elasticity gains more robust by improving residence time and targeting. Pairing HA scaffolds with peptides, antioxidants, or even regenerative approaches (e.g., growth-factor loading or stem-cell-friendly hydrogels) could further amplify ECM rebuilding while keeping the barrier comfortable. Longer, standardized clinical trials

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[51] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923)

[52] (Papakonstantinou, E., Roth, M., & Karakiulakis, G. (2012). Hyaluronic acid: A key molecule in skin aging. *Dermato-Endocrinology*, *4*(3), 253–258. https://doi.org/10.4161/derm.21923)

that track cellular markers (collagen, elastin, MMPs) alongside patient-reported outcomes will help compare platforms and guide best practices[53].

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[53] (Bravo, B., Correia, P., et al. (2022). Benefits of topical hyaluronic acid for skin quality and signs of skin aging: From literature review to clinical evidence. Dermatologic Therapy, 35(12), e15903. https://doi.org/10.1111/dth.15903)

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